

x-rite

PANTONE®

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日光



店内照明



家居照明

视觉评估工具

照明条件变化会引起颜色变化，这也是为什么受控光源对于颜色评估如此重要的原因。

爱色丽视觉评估解决方案采MacbethLighting技术，可以再现几乎所有照明场景，确保色彩在不同环境下始终准确一致。

- 色觉测试 FM100
- 光源箱
- 标准光源房

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全LED光源

挑战

欧盟关于禁用荧光照明的指令即将于2025年2月生效，要求各行业转向绿色LED解决方案。

中国政府也计划在2027年全面停止荧光灯的生产和使用。

影响

品牌商、供应商和制造商必须过渡至高性能的LED节能解决方案，才能保证跨不同材料的色彩一致性并符合监管要求。

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全LED光源

爱色丽解决方案

Judge LED光源箱是一种可靠准确的光源解决方案，支持7种LED光源，有效保证色彩一致性和合规性。

- **支持7种不同光源。**
- 客户可选：

D50, D65, UV, A,

LED-B2 3000K, LED-B3 4000K, 3500K

LED模拟荧光：CWF, TL84, U30, U35.

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	LED-B1	LED-B2	LED-B3	LED-B4	LED-B5
x	0,4560	0,4357	0,3756	0,3422	0,3118
y	0,4078	0,4012	0,3723	0,3502	0,3236
T_{cp}/K	2 733	2 998	4 103	5 109	6 598
Δ_{uv}	−0,0007	−0,0010	−0,0007	0,0005	0,0009
Special Colour Rendering Indices					
R_1	80	81	84	76	79
R_2	90	90	89	81	84
R_3	97	96	93	83	85
R_4	78	80	85	77	80
R_5	79	81	84	75	79
R_6	88	87	85	72	76
R_7	83	85	88	85	87
R_8	60	64	71	66	71
R_9	13	18	24	−1	7
R_{10}	78	77	74	53	60
R_{11}	75	77	84	72	77
R_{12}	71	68	66	47	50
R_{13}	82	83	85	76	81
R_{14}	99	98	96	90	91
General Colour Rendering Index					
R_a	82	83	85	77	80

CIE 015:2018

UDC: 535.66 535.67 612.843.31 159.937.51

Descriptor: Colorimetry; Colour of objects; Colour vision; Perception of colour

Illuminants LED-B1, LED-B2, LED-B3, LED-B4, and LED-B5 represent typical phosphor-converted blue LEDs at correlated colour temperatures, which are the most commonly used at the time of publishing this fourth edition of CIE 15. Illuminant LED-BH1 is representative for white LEDs that consist of a mixing of phosphor-converted blue LED and red LED (also called blue-hybrid LED). LED-RGB1 represents typical spectral shapes for mixing of red, green, and blue LEDs. LED-V1 and LED-V2 represent typical spectra shapes for phosphor-converted violet LEDs at two different correlated colour temperatures. However, LED-RGB1, LED-V1, and LED-V2 are not commonly used for white LED lamps at the time of publication of this report (Jost et al., 2017).

- **支持7种不同光源。**
- 客户可选：

D50, D65, UV, A,

LED-B2 3000K, LED-B3 4000K, 3500K

LED模拟荧光：CWF, TL84, U30, U35.

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实现可靠高效的色彩评估

- **准确复制**旧款CIE光源的照明效果，LED照明具提供了灯具过渡的一致性
- **开机即用，无需预热，**每天节省大量时间
- **稳定的光源，**保证色彩评估的一致性
- 经校准的**强度等级和稳定的色度参数**，避免明暗度不一致
- 经过精心设计的**光学扩散器**，能够在整个灯箱底部提供优质的光源均一性和漫射光效果



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用户至上：颜色评估功能的提升

- **用户友好型界面**配备**便捷按钮**，避免用户切换光源时分心
- **宽敞的观察区**适合各种样品尺寸
- **角度可调的观察支架**，以便从不同角度进行标准化样品对比 (选配)
- **远程控制功能**，用于轻松快速地切换光源 (选配)









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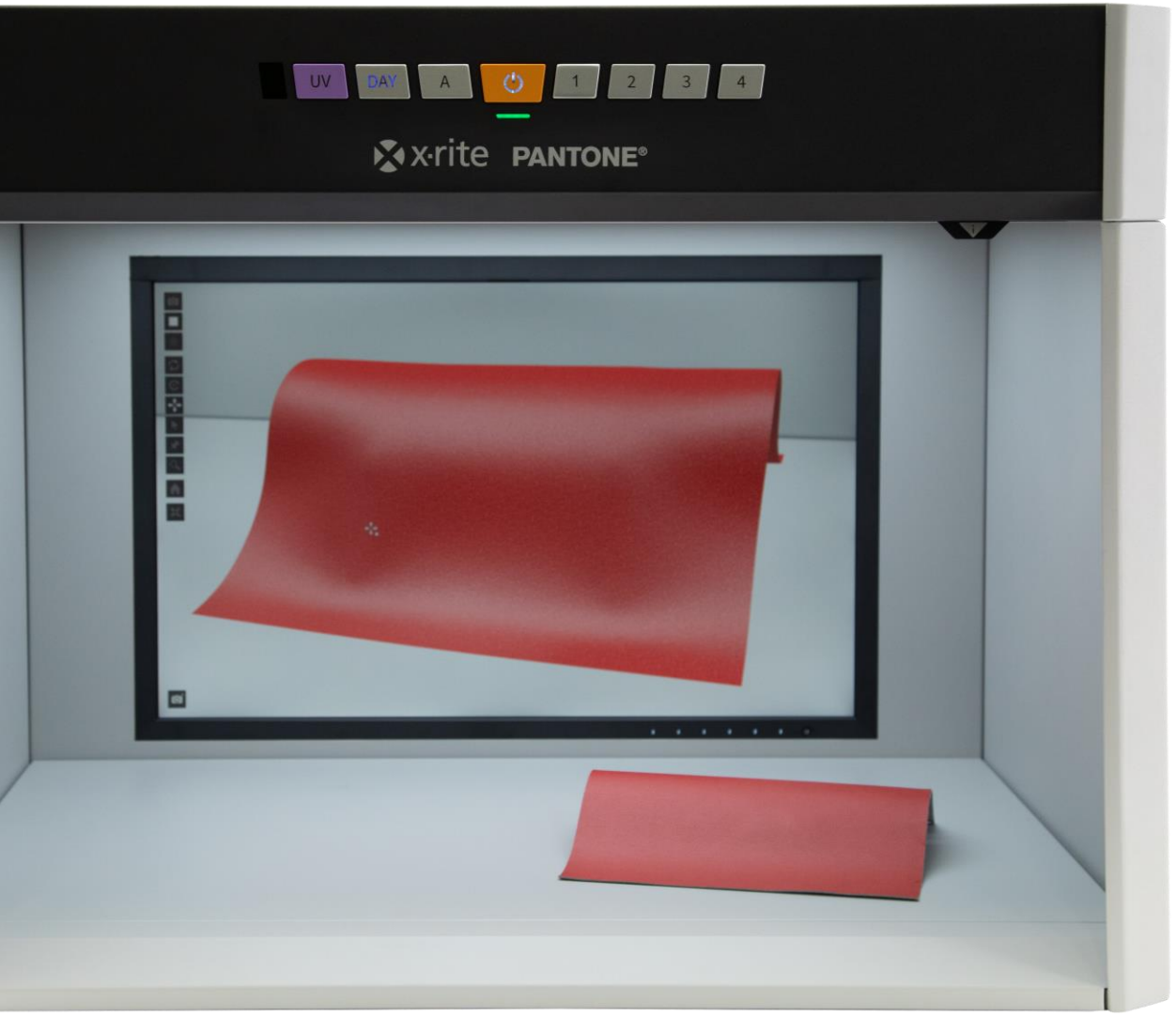


不只限于色彩视觉评估

- 定向D65日光光源可用于缺陷检测及橘皮效应等特殊特征的评估



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不只限于色彩视觉评估

- **内置专业校色显示器**可将实体样品与数字化图像进行对比。（选配）

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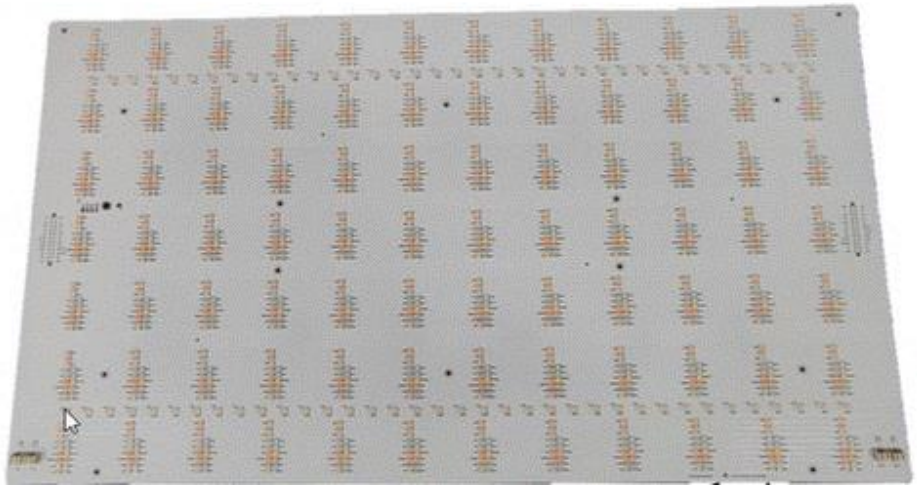
User-friendly UI with easy-to-press buttons

Avoid color assessment distractions with easy-to-press buttons



LED panel + Double optical diffuser

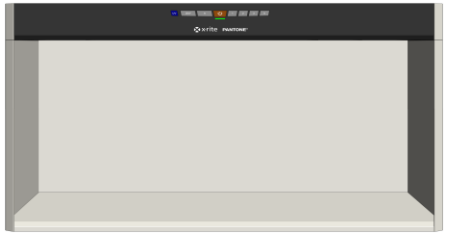
Great homogeneity and light diffusion



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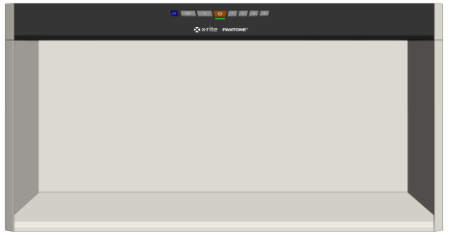


	JUDGE QC	JUDGE LED STD	JUDGE LED BIG	SPLQC
光源箱尺寸	D 53.5cm x W 68.5cm x H 54.5cm	D 53.5cm x W 71cm x H 54.5cm	D 75cm x W 150cm x H 75cm	D 61cm x W 94.5cm x H 95cm
价格（美金）	3000	4600	7900	9000
光源总数	5	7 + Dir D65	7 + Dir D65	7
LED技术光源	LED-B3, LED-B4	D65, D50, A, UV, LED-B2, LED-B3, 3500K, U30, U35, CWF, TL84	D65, D50, A, UV, LED-B2, LED-B3, 3500K, U30, U35, CWF, TL84	LED-B2, LED-B3, LED-B4
荧光技术光源	D65, D50, UV, U30, U35, CWF, TL84	-	-	UV, U30, U35, CWF, TL84
卤素技术光源	Inc A	-	-	D50, D65, Inc A, Hz
材质	钢	钢	钢	钢
箱体颜色	Munsell N7	Munsell N7	Munsell N7	Munsell N7
可见光同色异谱指数 (D65/D50)	B/C级	A/B级	A/B级	A/B级
紫外光同色异谱指数 (D65/D50)	C级	B/C级	B/C级	B级

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	JUDGE QC	JUDGE LED STD	JUDGE LED BIG	SPLQC
功率	~300W	~100W	~200W	~1150W
紫外灯	385nm	385nm	385nm	365nm
标准光源室（互联灯具）	×	√	√	√
灯具寿命	>2000 hrs.	>8500 hrs.	>8500 hrs.	>2000 hrs.
灯具计时器	全部	全部	全部	单独
调光荧光照明体	×	×	×	√
光学漫射器	×	√	√	√
均一性	>80%	>80%	>80%	>80%
与真实D65日光光谱的一致性	良	优	优	最优
爱色丽现场服务保养计划	×	√	√	√
包含/排除紫外光的D50/D65	×	√	√	√

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CERTIFICATE OF PERFORMANCE

x-rite PANTONE®

Judge LED® Viewing System

Serial No: *JLSBXRP-000010-0125-08*
Certification Date: *Jan 21 2025*
LightSpex Pro S/N: *2241664*
LightSpex Pro Cal. Date: *Jul 31 2024 (Due: Aug 31 2025)*
SKU Code: *JLSADGFCNSY*

Judge LED® is intended for use in visual color evaluation. This unit is tested by
measurement values for standard are traceable to National Institute of Standards
and test date pertaining to LightSpex Pro instruments are on file and are available f

Results

Light Source	Light Level Measured [lx]	Target [lx]	Color Temperature Measured [K]	Targe [K]
CIE D65	2080	2000 ± 200	6407	6500 ±
CIE D65+UV	2080	2000 ± 200	6421	6500 ±
CIE A	2037	2000 ± 200	2867	2856 ±
CIE D50	2037	2000 ± 200	5069	5000 ±
CIE D50+UV	2035	2000 ± 200	5078	5000 ±
U30 L	2017	2000 ± 200	2934	3000 ±
L40-B3	2079	2000 ± 200	4086	4103 ±
CWF L	2078	2000 ± 200	4309	4200 ±
GT D65	1929	1900 ± 300	6385	6350 ±
UVA	37 > 1.50 [µW/cm²/nm]		382 nm	385 ±

The results are from items tested only. Certification Ambient Temperature is 22-24
at 52 mm off the bottom of the light booth. The expanded uncertainty of the measu
(x,y) when viewing a 3000 K blackbody source. Illuminance levels are a result of d
effects of surrounds and maintenance.

***Date Placed in Operation:** _____ ****Due Date:** _____

X-Rite recommended certification interval: 12 months
*The "Date placed in Operation" should be established by the customer upon initial use of the unit.
**The "Due Date" for instrument recertification may be established (by the customer) by adding the "recommended certification interval" to the "Date Placed in Operation".
This is a computer-generated certificate, no signature is required.

This Performance Report shall not be reproduced except in full, without the approval of X-Rite Inc. This report relates to the performance of the lighting unit itself, and cannot be used to claim approval of products viewed since this process is subjective and varies from user to user. This report does not endorse the viewing process by NIST or any agency of the Federal Government. Judge LED® is a registered trademark of X-Rite Inc. This Certificate supersedes all previous Certificates of Performance.

X-Rite Incorporated ▸ 4300 44th Street SE ▸ Grand Rapids ▸ Michigan 49512 ▸ USA ▸ 800-248-9748 ▸ www.xrite.com

Results

Light Source	Light Level Measured [lx]	Target [lx]	Color Temperature Measured [K]	Target [K]	MI_{VIS}	MI_{UV}	Pass/Fail
CIE D65	2080	2000 ± 200	6407	6500 ± 300	0.15 [A]		pass
CIE D65+UV	2080	2000 ± 200	6421	6500 ± 300	0.17 [A]	0.98 [C]	pass
CIE A	2037	2000 ± 200	2867	2856 ± 100			pass
CIE D50	2037	2000 ± 200	5069	5000 ± 300	0.21 [A]		pass
CIE D50+UV	2035	2000 ± 200	5078	5000 ± 300	0.22 [A]	0.96 [C]	pass
U30 L	2017	2000 ± 200	2934	3000 ± 300			pass
L40-B3	2079	2000 ± 200	4086	4103 ± 200			pass
CWF L	2078	2000 ± 200	4309	4200 ± 400			pass
GT D65	1929	1900 ± 300	6385	6350 ± 300			pass
UVA	37 > 1.50 [µW/cm²/nm]		382 nm	385 ± 5 nm			pass

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•MIuv（紫外范围内的同色异谱指数）：

•是指在紫外光范围内，物体颜色在不同光源下表现出的颜色差异程度的指标。以标准 D65 灯管为例，其 MIuv<1.5，意味着在紫外光区域，该灯管照射下的物体颜色与在标准日光下的颜色差异较小，能更准确地还原物体在紫外光下的真实颜色，对于需要精确判断颜色在紫外光下表现的场景，如某些特殊材料的检测、荧光物质的观察等非常重要。

•MIvis（可见光范围内的同色异谱指数）：

•是衡量在可见光范围内，物体在不同光源下颜色匹配程度的参数。比如标准 D65 灯管的 MIvis<1.0，表明在可见光波段，该灯管能够使物体呈现出与在自然日光下高度相似的颜色，能让人们在观察物体颜色时获得更准确、可靠的视觉效果，在纺织、印染、涂料、塑料等对颜色视觉匹配要求高的行业应用广泛。

ISO 3664:2009

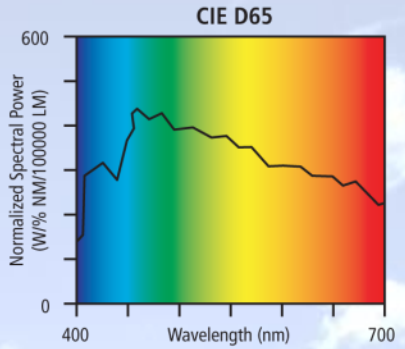
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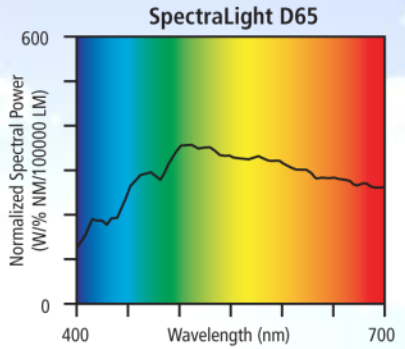
光源品质测量

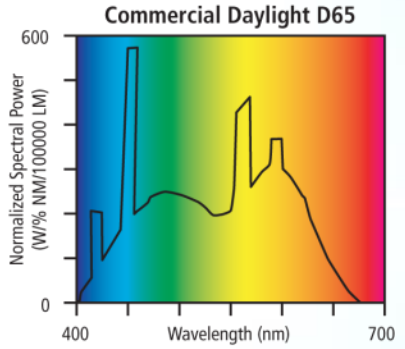
INTERNATIONAL COMMISSION ON ILLUMINATION

Publication 51 Daylight - 同色异谱指数Metamerism Index:
(现在为ISO/CIE SO12E)

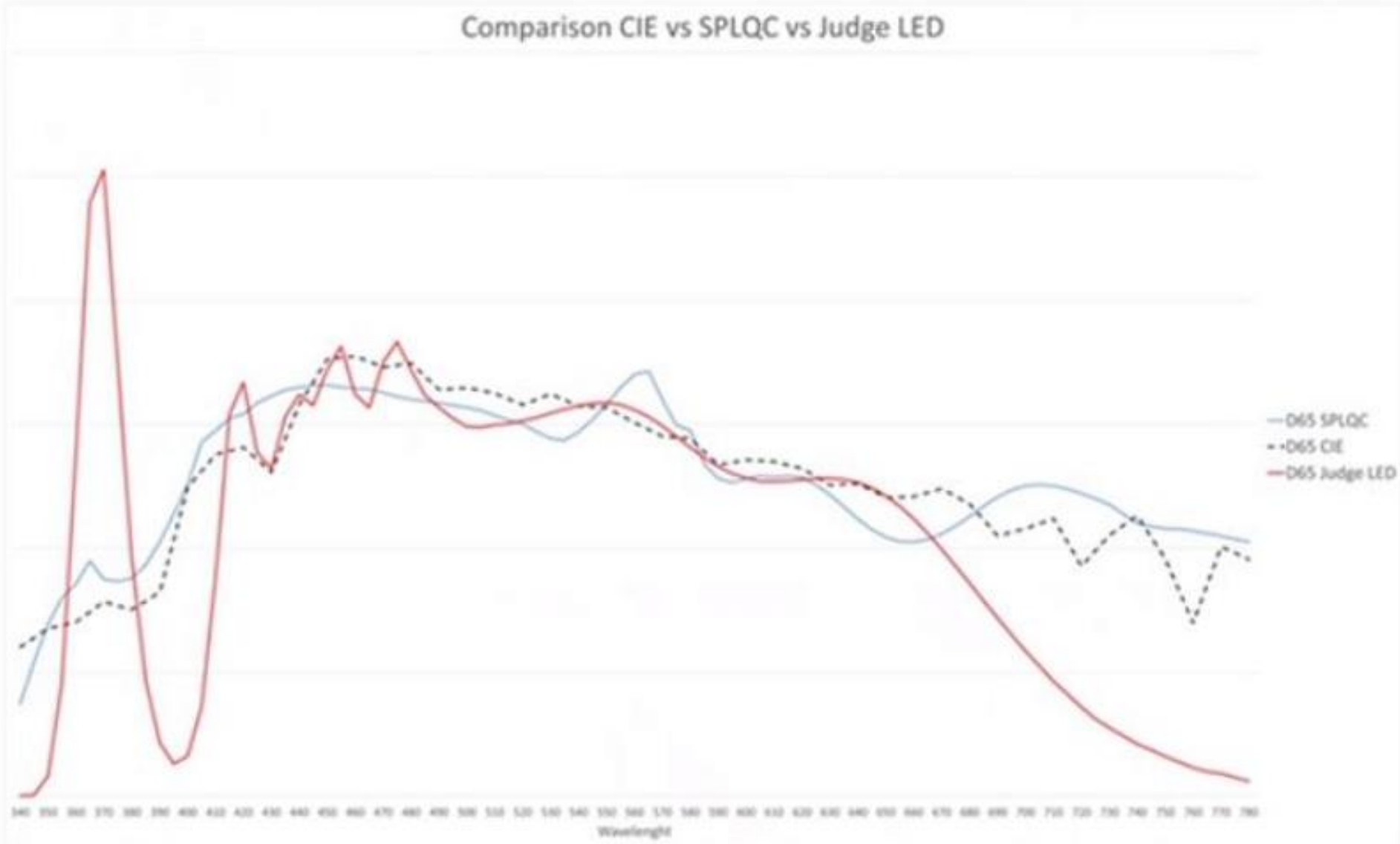
Quality Grade	Metamerism Index	
	CIELAB	CIELUV
A	<0.25	<0.32
B	0.25 to 0.50	0.32 to 0.65
C	0.50 to 1.00	0.65 to 1.30
D	1.00 to 2.00	1.30 to 2.60
E	>2.00	>2.60







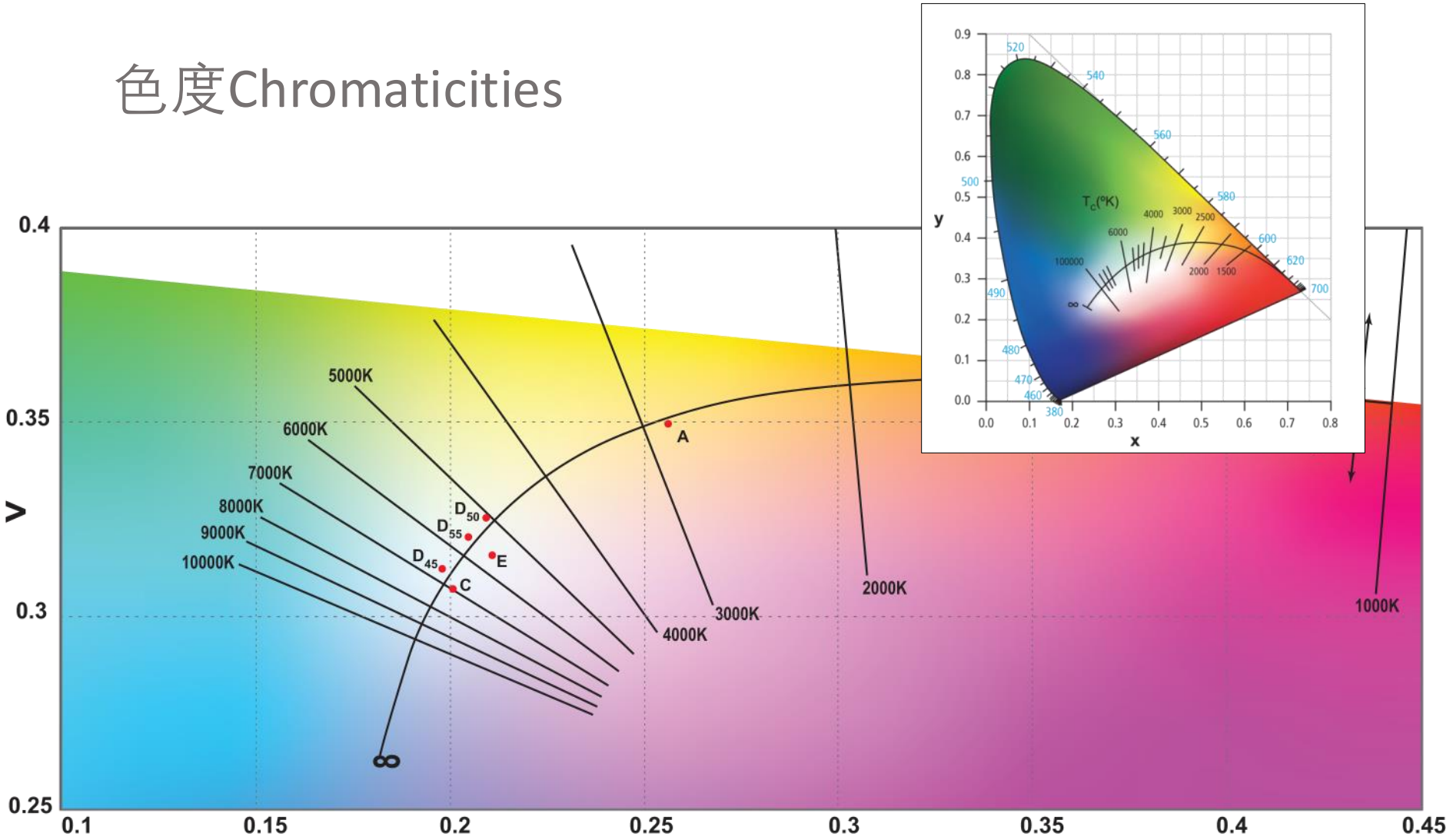
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光源品质测量

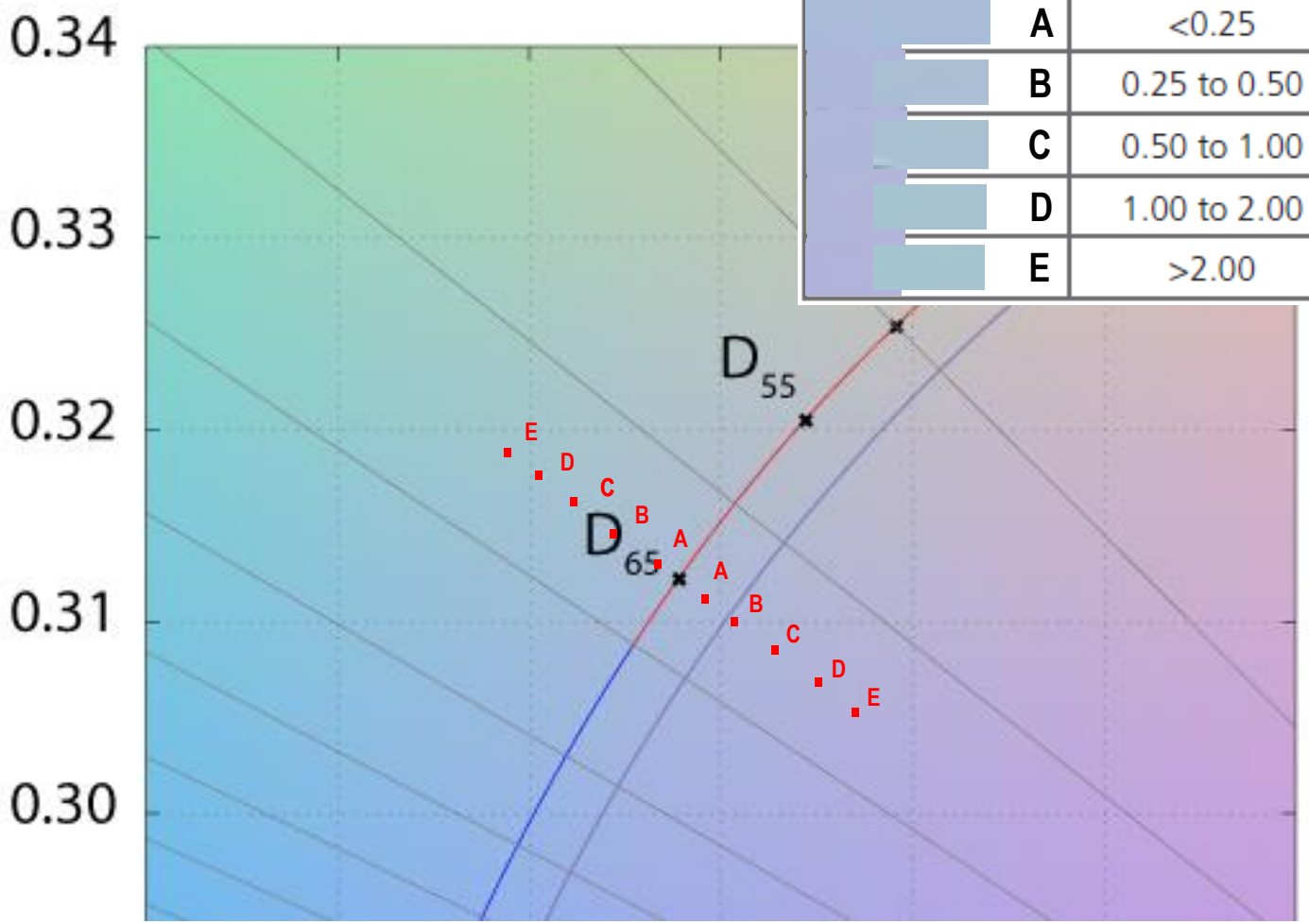
色度Chromaticities



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色度

Quality Grade	Metamerism Index	
	CIELAB	CIELUV
A	<0.25	<0.32
B	0.25 to 0.50	0.32 to 0.65
C	0.50 to 1.00	0.65 to 1.30
D	1.00 to 2.00	1.30 to 2.60
E	>2.00	>2.60



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纺织行业



Judge Qc

3000



Judge LED

4600



SPL Qc

9000

1



Ultraview

VeriVide

See in Truth

2



DCMB 2028

datacolor

3



P60 (6)

TILO

Price list:

~$7,000

~$5,000

<$1,000

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汽车行业



Judge Qc

3000



Judge LED

4600



SPL Qc

9000



byko-spectra pro

BYK



Color Control

Oligo surface controls



Color Control

JUST NORMLICHT

Price list: ~$8,000 ~$5,000 ~$12,000

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订单PN列表解释

Judge LED (D65/D50/U30/4000/CWF/A/UV)	JLSADGFCNSY	相当于 带有L840的Judge Qc双日光款
Judge LED (D65/U35/CWF/TL84/4000/A/UV)	JLSAHCIFNSY	相当于 带有L840的Judge Qc普通款
Judge LED (D65/D50/3000/3500/4000/A/UV)	JLSADBEFNSY	全新LED款,适合有JudgeQc的客户
Judge LED (D65/U35/3500/TL84/4000/A/UV)	JLSAHEIFNSY	同时做 沃尔玛、Jcpenny的客户
Judge LED (D65/U30/CWF/TL84/4000/A/UV)	JLSAGCIFNSY	相当于 带有L840的Judge Qc普通款

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消除色彩与外观之间的鸿沟

色彩是否准确一致关乎您的成败。

我们助您实现高水平的色彩再现。

